# MILITARY SPECIFICATION

# RANGE POT ASSEMBLY: 10553423

## 1. SCOPE

1.1 This specification covers one type range control known as Range Pot Assembly: 10553423. It is associated equipment but not an integral part of XM73 Reflex Sights and will be referred to herein as a range control.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

**SPECIFICATIONS**

Federal

| | |
|---|---|
| **PPP-T-60** | Tape, Pressure-Sensitive, Adhesive, Waterproof, For Packaging |

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Material: General Specification Governing the Manufacture and Inspection of |
| **MIL-P-13988** | Paper, Lens, Tissue, Antitarnish; Wrapping |
| **MIL-P-14232** | Parts, Equipment and Tools for Army Materiel: Packaging and Packing of |
| **MIL-I-45607** | inspection Equipment, Acquisition Maintenance and Disposition of |

**STANDARDS**

**Military**

| | |
|---|---|
| **MIL-STD-105** | Sampling Procedures and Tables for Inspection by Attributes |
| **MIL-STD-109** | Quality Assurance Terms and Definitions |

**DRAWINGS**

**U.S.** Army, Frankford Arsenal

| | |
|---|---|
| D10553423 | Range Pot Assembly: 10553423 |



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PACKAGING DATA SHEET

10553423 Packaging of Range Pot Assembly: 10553423

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

**3.1** Materials.- Materials shall be in accordance with drawings (see 3.2) material specifications and general specifications forming a part of this specification.

**3.2** Fabrication.- The range control shall be manufactured in accordance with Drawing D10553423 and all drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926 on the range control.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) part identification and marking
(e) Workmanship

**3.4** Environmental.

3.4.1 Storage at extreme temperatures.- The range control shall exhibit no evidence of damage after storage at -65° Fahrenheit (F) and at +160°F.

**3.4.2** Operability at extreme temperatures.- While thermally stabilized -at -40°F and at +150°F, the range control knob shall exhibit a running torque of one (1) to six (6) pound inches. Throughout the operational range, the range control knob shall be manually operable.

3.4.3 Humidity.- Exclusive of Connector MS3110-F-10-6P, the range control shall exhibit no evidence of damage subsequent to exposure to a 97% relative humidity (RH) for 24 hours at +160°F, followed by four (4) hours at between +60°F and +90°F with condensation taking place on the range control.

**3.4.4** Vibration (between +60°F and +90°F).- The range control shall not evidence physical damage or electrical circuit malfunction during vibration or subsequent to the two following vibration phases:



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a. **PHASE I.- Vibration in a vertical and lateral plane for 5 ± 0.5 minutes 'frequencies and amplitudes in Table I.**

**TABLE 1 - PHASE I VIBRATION**

| FREQUENCY (CPS) (BOTH PLANES) | DOUBLE AMPLITUDE (IN INCHES) (VERTICAL PLANE) | (LATERAL PLANE) |
|---|---|---|
| 5.4 | 0.128 | 0.066 |
| 11.9 | 0.038 | 0.032 |
| 47.5 | 0.008 | 0.006 |
| 65.0 | 0.006 | 0.004 |

b. **PHASE II.- Vibration in a vertical and lateral plane** varied continuously over a frequency range from 5 to 65 cps and return to 5 cps at a double amplitude of 0.072 inch or an acceleration of two G (s) whichever is lesser (see Figure 1) for a 5 ± 1.0 minute period in each plane.

**3.5** General (between (between +60°F and +90°F).

3 . 5 . 1 ~ . - The range control knob shall exhibit ease of manipulation while performing its intended function.

**3.5.2** Torque.- The range control knob shall exhibit a running torque of between one (1) and five (5) pound inches.

**3.5.3** Range control knob scale zero.- When the range control knob is in an extreme ccw position, its scale shall indicate zero within the width of a graduation line.

**3.5.4** Range control electrical resistance.- When the range control knob is in an extreme ccw position, the direct current (de) resistance between Pins A and B of Receptacle MS3112E-10-6P shall be between two and 20 ohms, and the dc resistance between A and C of the same receptacle shall be between 9,500 and 10,500 ohms. The dc resistance between Pins B and A shall increase when the range control knob is rotated clockwise. The dc resistance between Pins B and C shall be less than 10,000 ohms.

**3.5.5** interchangeability.- The range control shall meet the requirements of 3.4 and 3.5 without the need for selective components. Compatibility shall exist between mechanical interfaces of components required for range control maintainability and between the mechanical and electrical interfaces of any random combination of range controls and the interface of the next higher assembly.

## *4.* QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as



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otherwise specified, the supplier may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 **First article (initial Production) approval.- The requirement for first article approval and the responsibility (Government or contractor)** for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) range controls, plus three (3) each of all items covered by Supplementary Quality Assurance Provisions (SQAP) for the range control. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production

**4.2.1** Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

**4.2.2** contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

**4.3.1** Submission of product*- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP listed in the technical data package (TDP]. In the absence of SQAP'S, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examinations and tests related to Section 3 herein shall be performed on a single defect



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(individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables II and 111. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

| CLASS | | REQUIREMENT | TEST PROCEDURES |
|---|---|---|---|
| CRITICAL: | NONE | | |
| MAJOR : | AQL 1.0% Defective | | |
| 101. | vibration (PHASE II) | 3.4.4.(b) | 4.6.4 |
| 102. | Manual operability | 3.5.1 | 4.6.5.1 |
| 103 . | Torque | 3.5.2 | 4.6.5.2 |
| 104. | Range control knob scale zero | 3.5.3 | 4.6.5.3 |
| 105, | Range control electrical resistance | 3.5.4 | 4.6.5.4 |
| MINOR : | NONE | | |

NOTE : The tests in Table 11 shall be conducted in accordance with standard ambient test condition requirements in the "Test Facilities" requirement of MIL-F-13926.

**4.3.3** Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 Special sampling.- Samples shall be selected from lots at random using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. Range control samples shall have met all the requirements and tests outlined in Table II prior to and after being subjected to the tests in Table 111.

TABLE 111

| CHARACTERISTIC | | REQUIREMENT | TEST PROCEDURES |
|---|---|---|---|
| 301. | Vibration (PHASE 1) | 3.4.4(a) | 4.6.4 |
| 302. | Operability at extreme temperatures | 3.4.2 | 4.6.2 |
| 303 . | Interchangeability (Assembly only) | 3.5.5 | 4.6.5.5.2 |

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4.4.1.1 Failure of sample .- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3.).

**4.4.2 First article sample.- Unless Otherwise specified in the contract, test(s) listed herein shall be conducted on one unit of the initial production sample and shall not again be performed on the remaining production quantities, provided the manufacturing processes and techniques used to produce the samples have not been changed or revised.**

TABLE IV

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURES |
|---|---|---|
| **304.** Storage at extreme temperature | 3.4.1 | 4.6.1 |
| 305. Humidity | 3.4.3 | 4.6.3 |
| 306. lnterchangeability (Components and Assembly) | **3*5.5** | **4.6.5.5.1; 4.6.5.5.2** |

**4.4.3** Certification.- Certification shall include test data and results for all characteristics specified in Table V. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

**TABLE V**

| CHARACTERISTIC | **REQUIREMENT** | **TEST DATA TO COMPLY WITH** |
|---|---|---|
| **401.** Materials | 3.1 | Applicable drawings, standards, and specifications |
| **402.** Fabrication | 3.2 | |
| 403 . General specification (MIL-F-13926) | **3.3** | |
| 404. Interchangeability (Components only) | 3.5.5 | 4.6.5.5.1 |



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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

**4.5.2.2** Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Examination" and "Test Facilities\' requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90%. of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

**4.5.2.3** inspection equipment design requirements.- Inspection equipment provided in accordance with 4.5.2.2 shall meet the following requirements:

a. Environmental and mechanical test equipment.- Temperature, humidity, and vibration test equipment shall conform to applicable requirements in the "Test Facilities" requirement of MIL-F-13926. Standard vibration equipment provided shall have provisions for controlling the frequency and amplitude of vibration in accordance with 3.4.4. Vibration equipment shall have provisions for holding securely the range control throughout the vibration tests, and allow it to be orientated in a vertical and lateral plane for vibration. A standard time measuring instrument shall be provided to control vibration periods. Also provided shall be a standard torque wrench with appropriate adapters.

b. Electrical test equipment.- An electrical test circuit external to the range control shall be provided. When used in conjunction with an electrical power source, it shall provide the direct current (de) potential required by individual tests. It shall not be subject to the



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mechanical vibration of the vibration test, and it shall be connected electrically to items under tests by flexible conductors. This test circuit shall incorporate the required electrical instrumentation (see 4.5.2.4, Items 1 and 2) to measure and monitor specified electrical characteristics of the items under test. A low voltage ohmmeter (Item 3) also shall be provided to conduct electrical resistance measurements.

**4.5.2.4 Test equipment accuracy.- In conjunction With 4.5.2.3, test equipment shall have at least the accuracy depicted for the following equipment:**

| ITEM | NOMENCLATURE | DESCRIPTION |
|---|---|---|
| 1. | Voltmeter | 0-50 Vdc; full scale accuracy ± 2% |
| 2. | Ammeter ? | 0-l.0 Amp dc; full scale accuracy ***2%** |
| **3.** | **Ohmeter** | **Resistance ranges to indicate 20 ohms and 10,000 ohms mid-scale; midscale accuracy ± 3%** |

4.6 Test methods and procedures.

4.6.1 Storage at extreme temperatures.- Utilizing test equipment conforming to 4.5, gradually change chamber temperature to prevent thermal shock until the external surface of the range control is thermally stabilized at each extreme temperature specified in 3.4.1. Maintain each extreme temperature for an uninterrupted period of at least 24 hours. Upon completion of each exposure the temperature shall be gradually returned to between +60°F and +90°F. Subsequent to completion of each temperature excursion, the range control shall be examined to determine that no evidence of damage exists.

**4.6.2** Operability at extreme temperatures.- Utilizing test equipment conforming to 4.5, gradually change chamber temperature to prevent thermal shock and until the range control is thermally stabilized at temperatures in 3.4.2. Maintain extreme temperatures for only the uninterrupted period of time required to perform the test and examinations specified herein. The test and examination in 4.6.5.1 shall be performed to verify manual operability of the range control at each extreme temperature in compliance with 3.4.2. Utilize the same test procedure specified in 4.6.5.2 and determine that running torque of the elevation knob and range control knob is within the limits in 3.4.2.

4.6.3 Humidity.- Utilize test equipment conforming to 4.5 and perform this test in accordance with Procedure III of the humidity test in MIL-F-13926, except that the range control shall be exposed for at least 24 hours to a 97% relative humidity (RH) at an ambient temperature of +160°F. Chamber temperature shall then be reduced to between +60°F and +90°F with condensation and the range control exposed *to* this condition for at least four (4) hours, within which time chamber temperature may be adjusted to



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obtain other condensation periods. The humidity phase of this test shall now cease and the external surfaces of the range control shall be dried by wiping all accessible areas. Dyring by air blast or in an oven shall not be permitted.

"CAUTION: Performance requirements of this specification require that specified items be protected from direct exposure to this humidity test. Such items, as are specified, shall first be wrapped with paper conforming to MIL-P-13988 and then completely covered with adhesive paper tape conforming to PPP-T-60. Subsequent to this test, remove the protective covering and verify that all previously covered surfaces are free of foreign matter." "

4.6.4 Vibration test requirements.- Perform vibration tests under the standard ambient test conditions in the "Test Facilities" requirement of MIL-F-13926, utilizing *test* equipment conforming to 4.5 in accordance with the requirements and examinations specified herein. Secure the range control to the vibration table making all electrical connections in proper polarity, and apply a direct current (de) potential to the adjusted range control through its electrical input connector. Adjust the range control knob to establish approximately equal electrical resistance between stops - with respect to the potentiometer wiper terminal - and apply sufficient dc potential to cause each half of the potentiometer winding to draw at least three (3) milliamperes (dc) but in no case more than six (6) milliamperes (de). Throughout vibration, electrical circuits shall be under constant surveillance. Monitor all electrical circuits to determine that no input current change occurs indicative of an intermittent electrical connection or short circuit. Upon completion of either vibration; or both, the range control shall be examined to determine that no evidence of damage exists.

4.6.5 General performance test requirements.- Tests herein shall be performed under the standard ambient test conditions in the "Test Facilities" requirement of MIL-F-13926, utilizing test equipment conforming to 4.5.

4.6.5.1 Manual operability.- The range control shall be manually rotated throughout its entire rotational range - from stop to stop - in a cw and ccw direction to determine ease of manual operation with no evidence of binding or slip, Also observe during this examination that the knob is secure to its shaft and that compatibility exists between the control's intended function and associated markings.

4.6.5.2 Torque.- This test shall be performed by applying a force normal to the range control shaft. After breakaway, the range control knob shall be rotated cw and ccw through its entire excursion and the torque required to maintain rotation noted. These torque values shall be within the ranges specified in 3.5.2.

4.6.5.3 Range control knob scale zero. - Examine the range knob scale to determine alignment between its numeral zero and the index on Bracket 10553430 in compliance with 3.5.3.



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4.6.5.4 Range control electrical resistance.- Utilizing low voltage ohmeter as specified in 4.5.2.4, conduct this test at the output pins of the range control assembly's electrical connector. Select the proper connector pins and measure the specified values of electrical resistance to determine compliance with 3.5.4. Also observe a progressive linear increase and decrease in electrical resistance during rotation of the range control knob. Utilize three test leads - two of which shall be common - when conducting this test and make test lead connections in such a way as to detect any evidence of circuit malfunction.

4.6.5.5 Interchangeability.

**4.6.5.5.1** Components.- Perform this test subsequent to meeting all other requirements in Section 3 by disassembling and reassembling each range control as follows:

a. Divide range control components into three groups of nominated components, as depicted in Table VI-1, and distribute the groups into three separate areas, as shown in Table **VI-2, until each area contains components for** a complete range control.

b. Take in order, groups of components for the first range control and place in Areas A, B and C, respectively, and likewise: the second in B, C and A, and the third in C, A and B.

*c. Reassemble range controls using only components which are in the same area and verify that each reassembled range control conforms to requirements 3.5.1 to 3.5.4 inclusive.

TABLE VI-1: GROUPS OF NOMINATED COMPONENTS

| (GROUP 1) | (GROUP 2) | (GROUP 3) |
|---|---|---|
| **10553530** | 10553429 | 10553428 |
| **10553419** | 10553348 | 10553347 |

Components not listed in
Groups 1, 2 and 3, herein

TABLE VI-2: COMPONENT DISBURSEMENT AFTER INTERCHANGE

| | (AREA "A") | (AREA "B") | (AREA "C") |
|---|---|---|---|
| RANGE CONTROL #1 | GROUP 1 | GROUP 2 | GROUP 3 |
| #2 | 2 | 3 | 1 |
| #3 | 3 | 1 | 2 |

*Commercial components such as screws, nuts, washers, pins or packing which are rendered unserviceable by disassembly shall be replaced without penalty to this test.



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**4.6.5.5.2 Assembly.- Examine Bracket 10553430 for compliance with dimensions and tolerances on Drawing C10553430, which are applicable to the mechanical interface between the completely assembled range control** and its next higher assembly, to verify compliance with 3.5.5.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- packaging, packing and marking shall be in accordance with Packaging Data Sheet 10553423. The level of protection shall be as specified in the procurement document.

## 6. NOTES

6.1 Intended use.- When mounted with the XM28 or TAT 102 Armament Subsystem, the range control is used to aim the 7.62mm gun and/or 40mm grenade launcher in the turret from the pilots position of the AH-lG Helicopter. The electrical power requirements are supplied by the aircraft's electrical system. Although not an integral part of each type. of XM73 Reflex Sight, the range control is provided with the sight for elevation and depression of the minigun that is a part of the XM28 Armament Subsystem.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable packaging data sheet number (see 5.1)
(d) Applicable stock number

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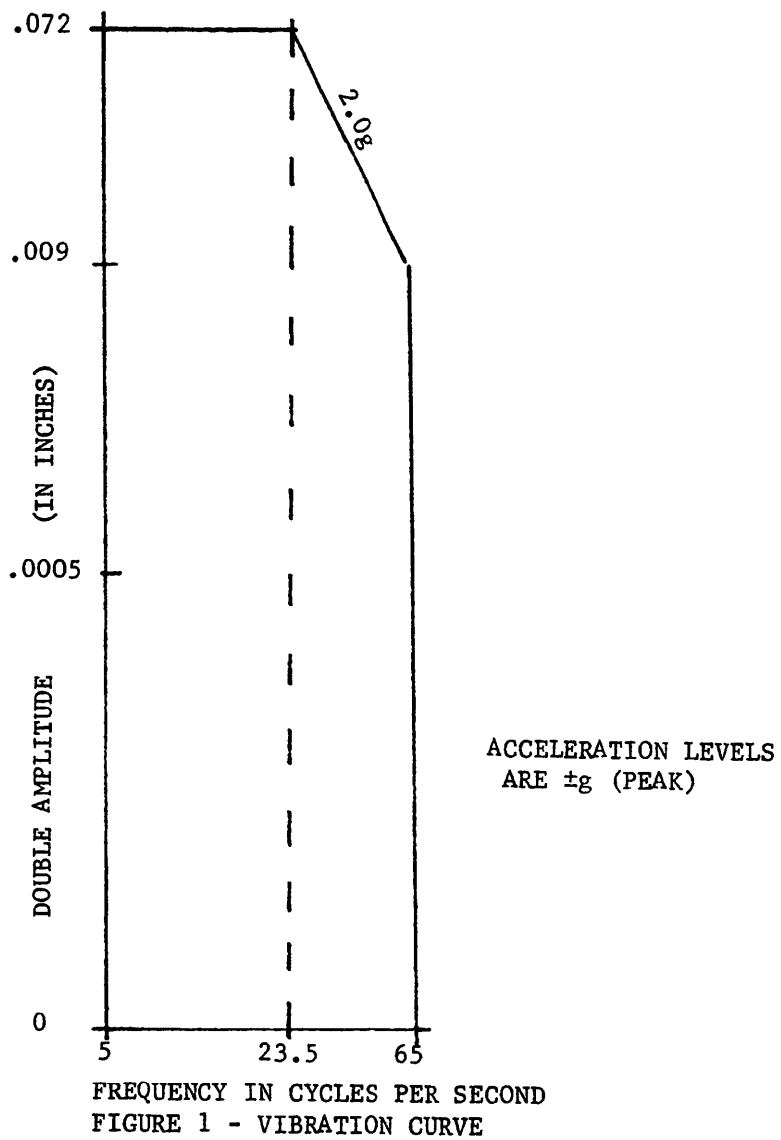



FIGURE 1 - VIBRATION CURVE



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# SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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